Operating system

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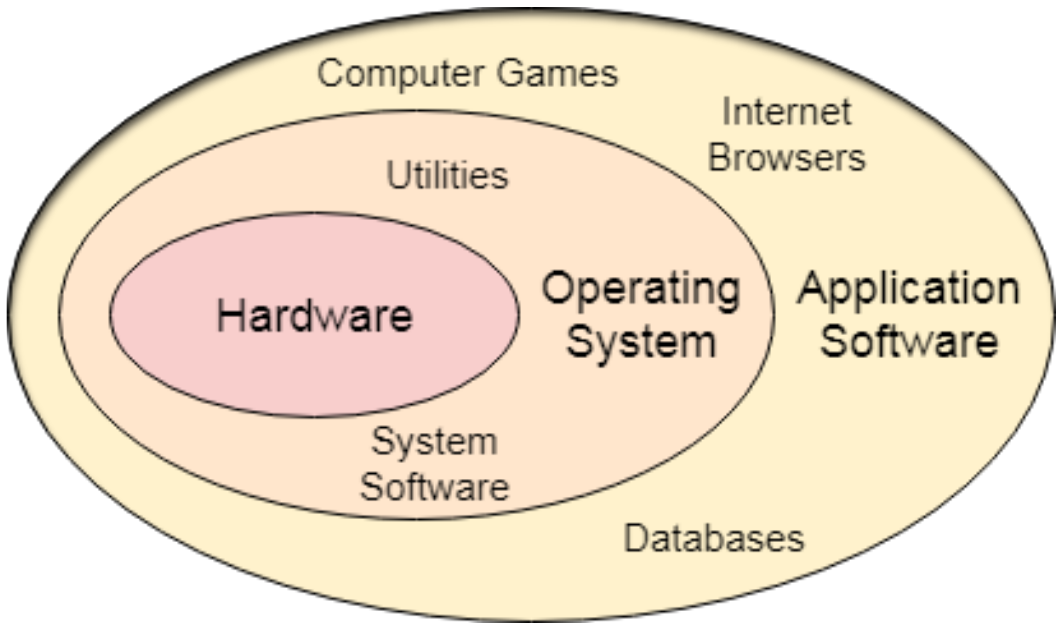
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What is an operating system?



Definition: An **operating system** is the **most important software** that runs on a computer. It manages the computer's **memory** and **processes**, as well as all of its **software** and **hardware**. It also allows you to **communicate** with the computer without knowing how to speak the computer's language. **Without an operating system, a computer is useless**.

Definition: An **Operating System** can be defined as an **interface between user and hardware**. It is responsible for the execution of all the processes, Resource Allocation, CPU management, File Management and many other tasks.

The purpose of an operating system is to provide an environment in which a user can execute programs in convenient and efficient manner.

What is an operating system?

- Execute user programs and make solving user problems easier
- Make the computer system convenient to use
- Use the computer hardware in an efficient manner

Goal’s of operating system?

The three most common operating systems for personal computers are

- **Microsoft Windows**,
- **macOS**, and
- **Linux**.

Modern operating systems use a **graphical user interface**, or **GUI**. A GUI lets you use your mouse to click **icons**, **buttons**, and **menus**, and everything is clearly displayed on the screen using a combination of **graphics** and **text**.

Types of operating system

Microsoft created the **Windows** operating system in the mid-1980s. There have been many different versions of Windows, but the most recent ones are **Windows 10** (released in 2015), **Windows 8** (2012), **Windows 7** (2009), and **Windows Vista** (2007). Windows comes **pre-loaded** on most new PCs, which helps to make it the **most popular operating system** in the world.

Microsoft Windows

macOS (previously called **OS X**) is a line of operating systems created by Apple. It comes preloaded on all Macintosh computers, or Macs. Some of the specific versions include **Mojave** (released in 2018), **High Sierra** (2017), and **Sierra** (2016).

According to **StatCounter Global Stats**, macOS users account for less than **10%** of global operating systems—much lower than the percentage of Windows users (more than **80%**). One reason for this is that Apple computers tend to be more expensive. However, many people do prefer the look and feel of macOS over Windows.



Linux (pronounced **LINN-ux**) is a family of **open-source** operating systems, which means they can be modified and distributed by anyone around the world. This is different from **proprietary software** like Windows, which can only be modified by the company that owns it. The advantages of Linux are that it is **free**, and there are many different **distributions**—or versions—you can choose from.

According to **StatCounter Global Stats**, Linux users account for less than **2%** of global operating systems. However, most **servers** run Linux because it's relatively easy to customize.

Linux

The operating systems we've been talking about so far were designed to run on **desktop** and **laptop** computers. **Mobile devices** such as **phones**, **tablet computers**, and **MP3 players** are different from desktop and laptop computers, so they run operating systems that are designed specifically for mobile devices. Examples of mobile operating systems include **Apple iOS** and **Google Android**.

Operating systems for mobile devices

Computer system can be divided into four components

- **Hardware** – provides basic computing resources, ex. CPU, memory, I/O devices
- **Operating system** - Controls and coordinates use of hardware among various applications and users
- **Application programs** – define the ways in which the system resources are used to solve the computing problems of the users, ex. Word processors, compilers, web browsers, database systems, video games
- **Users** - People, machines, other computers

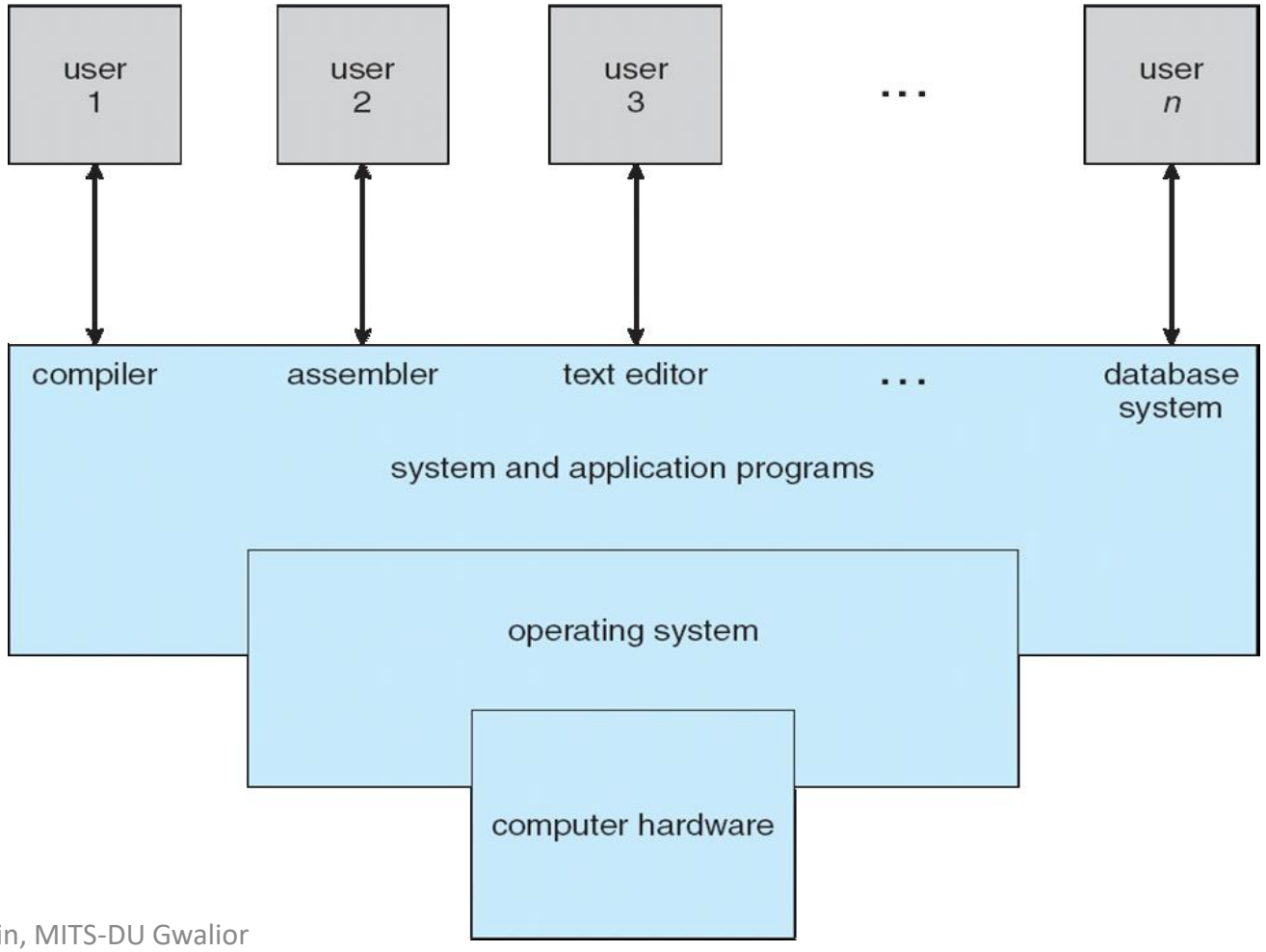
Computer System Structure

Components of a Computer System

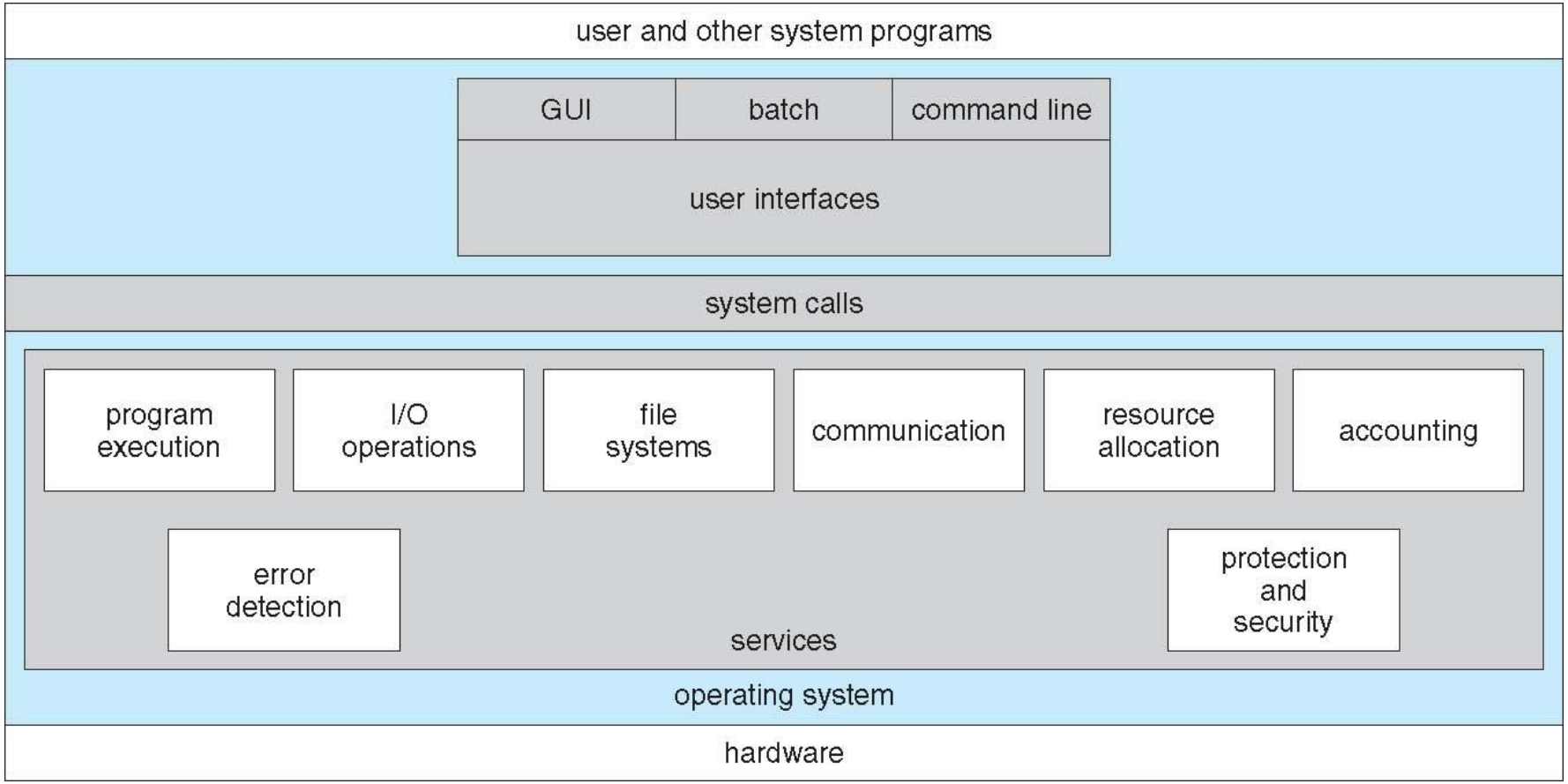
Components of OS: OS has two parts.

(1) **Kernel** is an active part of an OS i.e., it is the part of OS running at all times. It is a program which can interact with the hardware. Ex: Device driver, .dll files, system files etc.

(2) **Shell** is called as the command interpreter. It is a set of programs used to interact with the application programs. It is responsible for execution of instructions given to OS (called commands).



A View of Operating System Services



Operating System Services

Following are the five services provided by operating systems to the convenience of the users.

1. Program Execution

The purpose of computer systems is to allow the user to execute programs. So the operating system provides an environment where the user can conveniently run programs. Running a program involves the allocating and deallocating memory, CPU scheduling in case of multiprocessing.

2. I/O Operations

Each program requires an input and produces output. This involves the use of I/O. So the operating systems are providing I/O makes it convenient for the users to run programs.

3. File System Manipulation

The output of a program may need to be written into new files or input taken from some files. The operating system provides this service.

Operating System Services...

4. Communications

The processes need to communicate with each other to exchange information during execution. It may be between processes running on the same computer or running on the different computers. Communications can occur in two ways: (i) shared memory or (ii) message passing

5. Error Detection

An error is one part of the system may cause malfunctioning of the complete system. To avoid such a situation operating system constantly monitors the system for detecting the errors. This relieves the user of the worry of errors propagating to various part of the system and causing malfunctioning.

Operating System Services

Following are the three services provided by operating systems for ensuring the efficient operation of the system itself.

1. Resource allocation

When multiple users are logged on the system or multiple jobs are running at the same time, resources must be allocated to each of them. Many different types of resources are managed by the operating system.

2. Accounting

The operating systems keep track of which users use how many and which kinds of computer resources. This record keeping may be used for accounting (so that users can be billed) or simply for accumulating usage statistics.

3. Protection

When several disjointed processes execute concurrently, it should not be possible for one process to interfere with the others, or with the operating system itself. Protection involves ensuring that all access to system resources is controlled. Security of the system from outsiders is also important. Such security starts with each user having to authenticate him to the system, usually by means of a password, to be allowed access to the resources.

1. Process Management
2. Process Synchronization
3. Memory Management
4. CPU Scheduling
5. File Management
6. Security

Race Condition

A race condition occurs when multiple processes or threads access a shared resource simultaneously, and the final outcome depends on the execution order, leading to unpredictable behavior.

What does an Operating system do?

Process Management

A *process* is a program in execution. A process needs certain resources, including CPU time, memory, files, and I/O devices, to accomplish its task.

The operating system is responsible for the following activities in connection with process management.

- ✦ Process creation and deletion.
- ✦ process suspension and resumption.
- ✦ Provision of mechanisms for:
  - process synchronization
  - process communication

Process Synchronization
Ensures that multiple processes run smoothly without conflicts.
Uses synchronization mechanisms like semaphores and mutex locks.
Prevents race conditions (when multiple processes access shared resources).
Manages inter-process communication (IPC).

Process Management
Handles creation, execution, and termination of processes.
Manages process scheduling to optimize CPU usage.
Allocates system resources to processes.
Maintains process control blocks (PCB) to track execution state.

Main-Memory Management

Memory Management
Allocates and deallocates memory to processes dynamically.
Uses techniques like paging, segmentation, and virtual memory.
Ensures memory protection to prevent unauthorized access.
Keeps track of free and used memory areas.

Memory is a large array of words or bytes, each with its own address. It is a repository of quickly accessible data shared by the CPU and I/O devices. Main memory is a volatile storage device. It loses its contents in the case of system failure.

The operating system is responsible for the following activities in connections with memory management:

- ♦ Keep track of which parts of memory are currently being used and by whom.
- ♦ Decide which processes to load when memory space becomes available.
- ♦ Allocate and de-allocate memory space as needed.

File Management

File Management
Organizes and manages files on storage devices.
Provides a hierarchical directory structure.
Handles file operations like reading, writing, and deleting.
Enforces access control and permissions.

A file is a collection of related information defined by its creator. Commonly, files represent programs (both source and object forms) and data.

The operating system is responsible for the following activities in connections with file management:

- ✦ File creation and deletion.
- ✦ Directory creation and deletion.
- ✦ Support of primitives for manipulating files and directories.
- ✦ Mapping files onto secondary storage.
- ✦ File backup on stable (non-volatile) storage media.

I/O System Management

The I/O system consists of:

- A buffer-caching system
- A general device-driver interface
- Drivers for specific hardware devices

Secondary Storage Management

Since main memory (*primary storage*) is volatile and too small to accommodate all data and programs permanently, the computer system must provide *secondary storage* to back up main memory.

Most modern computer systems use disks as the principle on-line storage medium, for both programs and data.

The operating system is responsible for the following activities in connection with disk management:

- ✦ Free space management
- ✦ Storage allocation
- ✦ Disk scheduling

Distributed Systems (Networking)

A *distributed* system is a collection processors that do not share memory or a clock. Each processor has its own local memory.

- The processors in the system are connected through a communication network.
- Communication takes place using a *protocol*.
- A distributed system provides user access to various system resources.
- Access to a shared resource allows:
  - Computation speed-up
  - Increased data availability
  - Enhanced reliability

ex. internet, cloud coumputing, blockchain ,distributed databases

Protection System

Protection refers to a mechanism for controlling access by programs, processes, or users to both system and user resources.

The protection mechanism must:

- ✦ distinguish between authorized and unauthorized usage.
- ✦ specify the controls to be imposed.
- ✦ provide a means of enforcement.

A Protection System in an operating system (OS) ensures that processes, users, and system resources are safeguarded from unauthorized access, modification, or damage. It prevents security breaches, data corruption, and system failures by enforcing access control mechanisms.

Command Interpreter System

Many commands are given to the operating system by control statements which deal with:

- ✦ process creation and management
- ✦ I/O handling
- ✦ secondary-storage management
- ✦ main-memory management
- ✦ file-system access
- ✦ protection
- ✦ networking

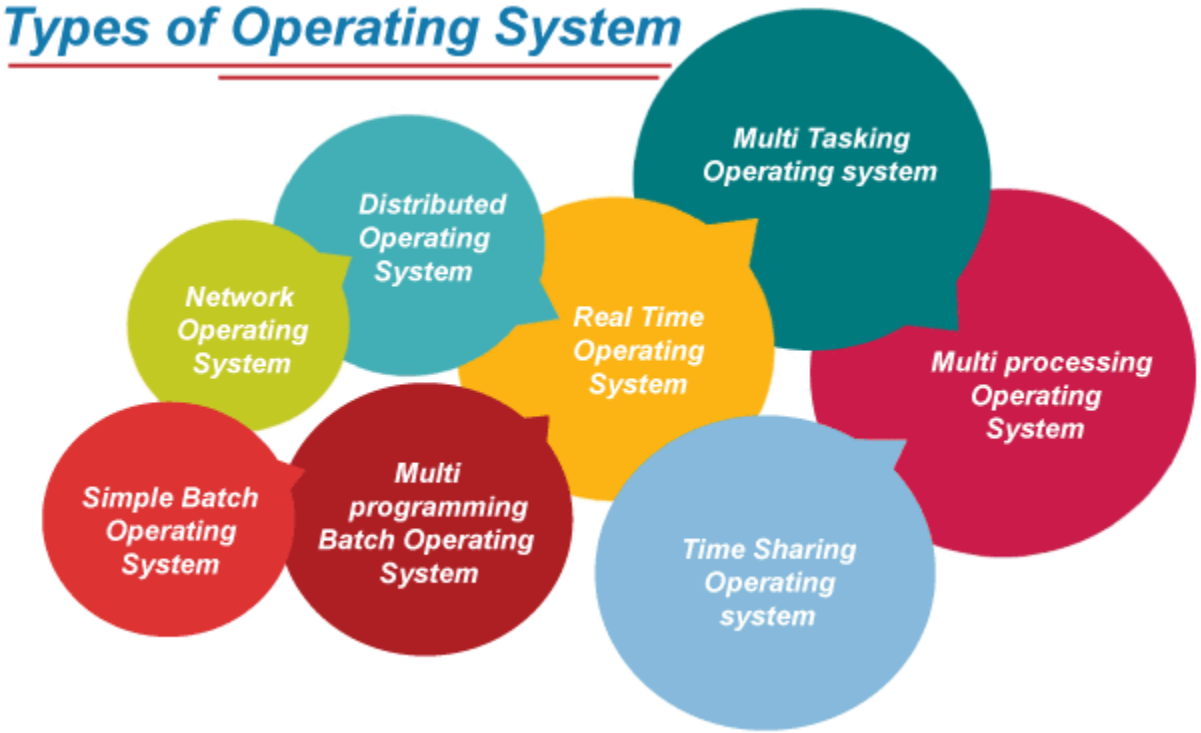
A Command Interpreter System is a part of the Operating System (OS) that processes and executes user commands. It acts as an interface between the user and the OS kernel, allowing users to interact with the system using text-based commands.

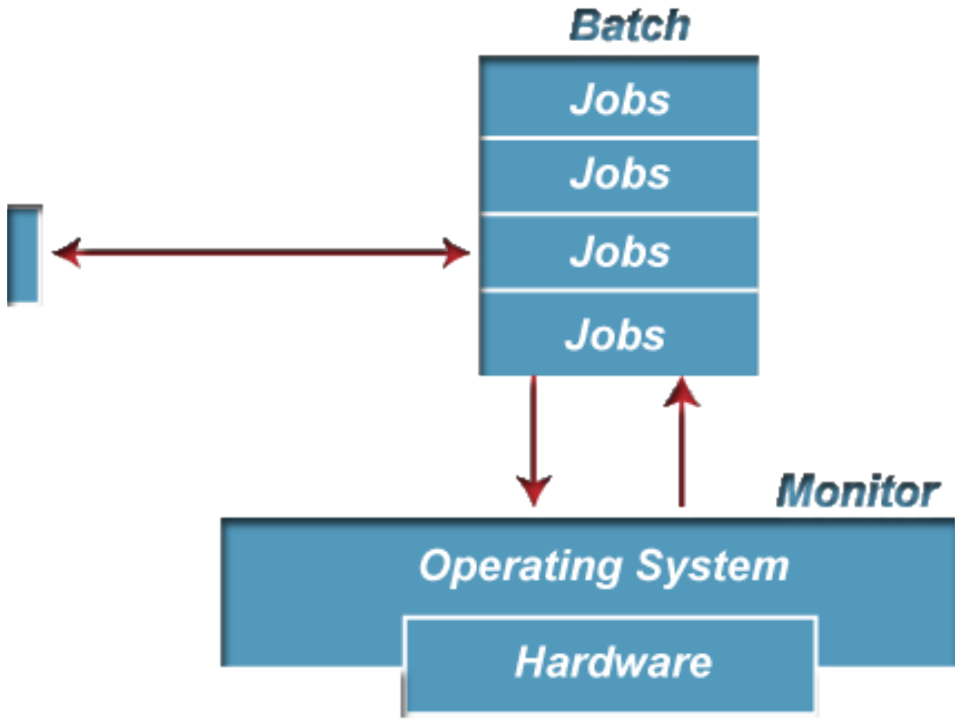
The program that reads and interprets control statements is called variously:

- ✦ command-line interpreter
- ✦ shell (in UNIX) : Its function is to get and execute the next command statement.

Types of Operating Systems

An operating system is a well-organized collection of programs that manages the computer hardware. It is a type of system software that is responsible for the smooth functioning of the computer system.





Batch Operating System

- In the 1970s, Batch processing was very popular. In this technique, similar types of jobs were batched together and executed in time. People were used to having a single computer which was called a mainframe.
- The system put all of the jobs in a queue on the basis of first come first serve and then executes the jobs one by one. The users collect their respective output when all the jobs get executed.

Advantages of Batch OS

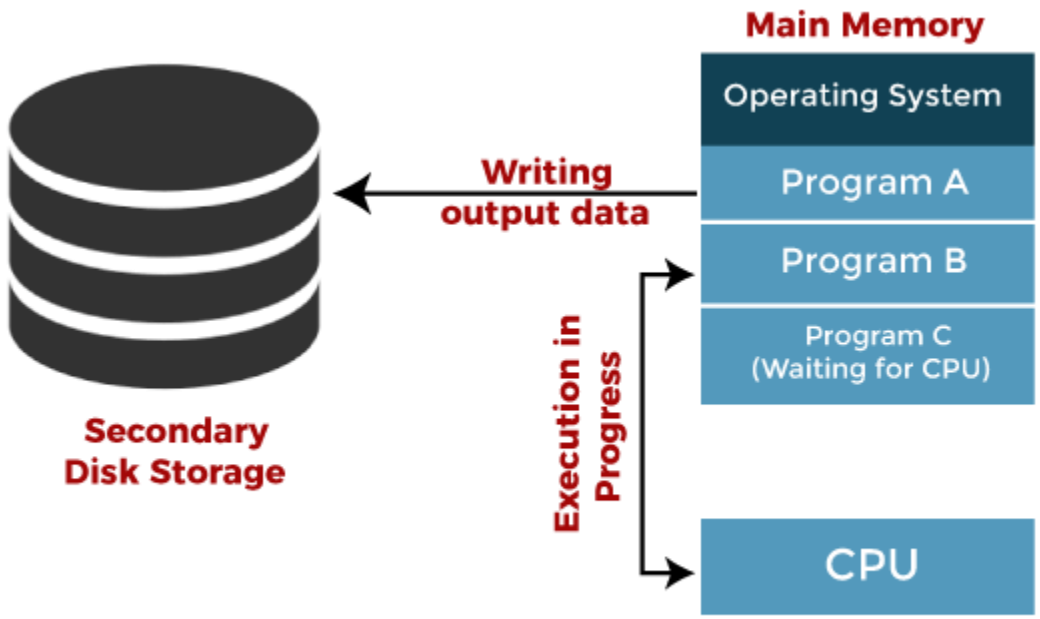
- The use of a resident monitor improves computer efficiency as it eliminates CPU time between two jobs.

Disadvantages of Batch OS

1. Starvation : Batch processing suffers from starvation.

2. Not Interactive : Batch Processing is not suitable for jobs that are dependent on the user's input. If a job requires the input of two numbers from the console, then it will never get it in the batch processing scenario since the user is not present at the time of execution.

Multiprogramming Operating System



Jobs in multiprogramming system

What Happens After the I/O is Completed?
Once the I/O operation is finished, the job does not immediately resume execution.
It is placed back into the ready queue and waits for the CPU.
The CPU continues executing the process that was loaded during the I/O wait.
When the scheduler selects the job again, it resumes execution from where it left off.

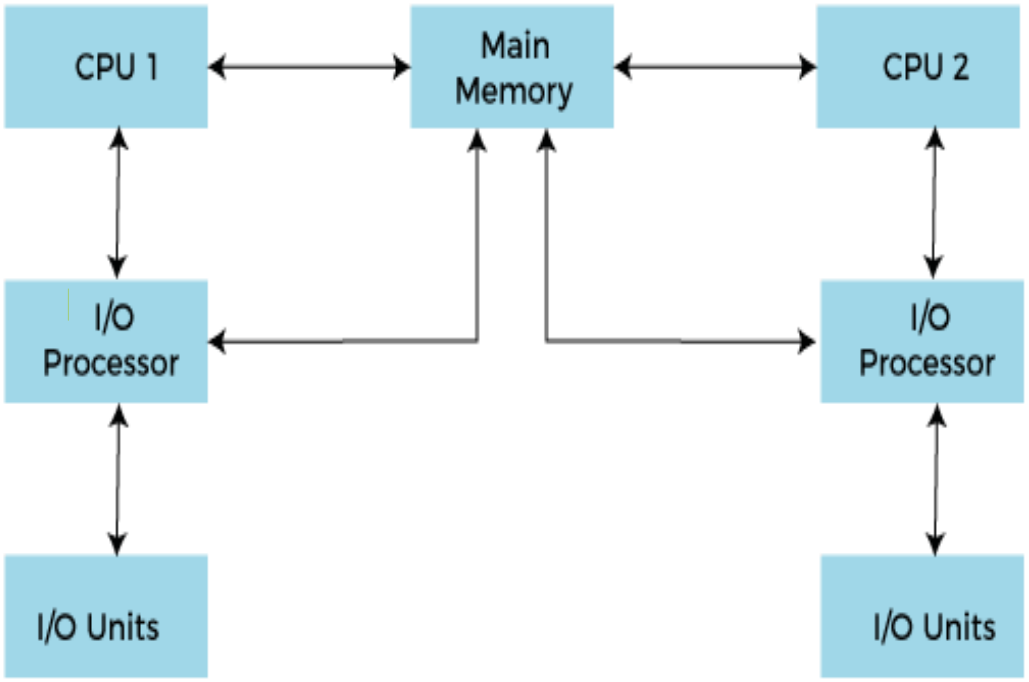
- Multiprogramming is an extension to batch processing where the CPU is always kept busy. Each process needs two types of system time: CPU time and IO time.
- In a multiprogramming environment, when a process does its I/O, The CPU can start the execution of other processes. Therefore, multiprogramming improves the efficiency of the system.

Advantages of Multiprogramming OS

- Throughout the system, it increased as the CPU always had one program to execute.
- Response time can also be reduced.

Disadvantages of Multiprogramming OS

- Multiprogramming systems provide an environment in which various systems resources are used efficiently, but they do not provide any user interaction with the computer system.



Working of Multiprocessor System

In Multiprocessing, Parallel computing is achieved. There are more than one processors present in the system which can execute more than one process at the same time. This will increase the throughput of the system.

- Types of multiprocessing system are
  - Symmetrical multiprocessing operating system
  - Asymmetrical multiprocessing operating system

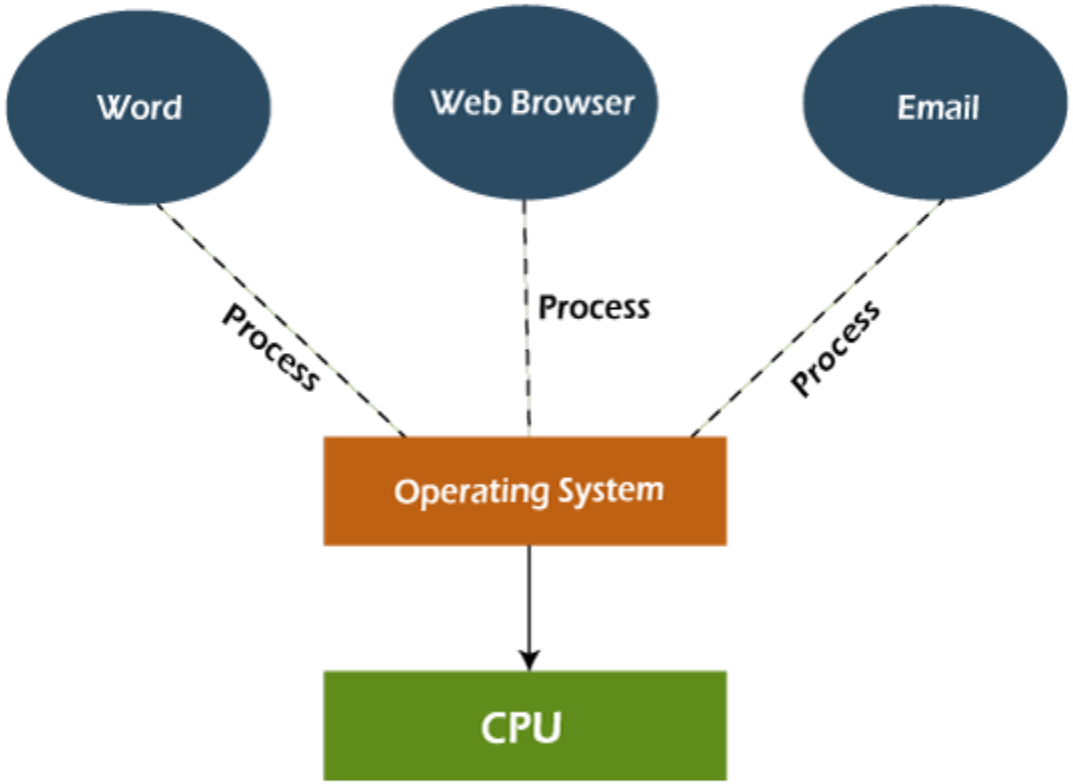
Multiprocessing Operating System

Advantages of Multiprocessing OS

- **Increased reliability:** Due to the multiprocessing system, processing tasks can be distributed among several processors. This increases reliability as if one processor fails, the task can be given to another processor for completion.
- **Increased throughout:** As several processors increase, more work can be done in less.

Disadvantages of Multiprocessing OS

- Multiprocessing operating system is more complex and sophisticated as it takes care of multiple CPUs simultaneously.



The multitasking operating system is a logical extension of a multiprogramming system that enables **multiple** programs simultaneously. It allows a user to perform more than one computer task at the same time.

Types of multitasking system are

- Pre-emptive Multitasking operating system
- Cooperative Multitasking operating system

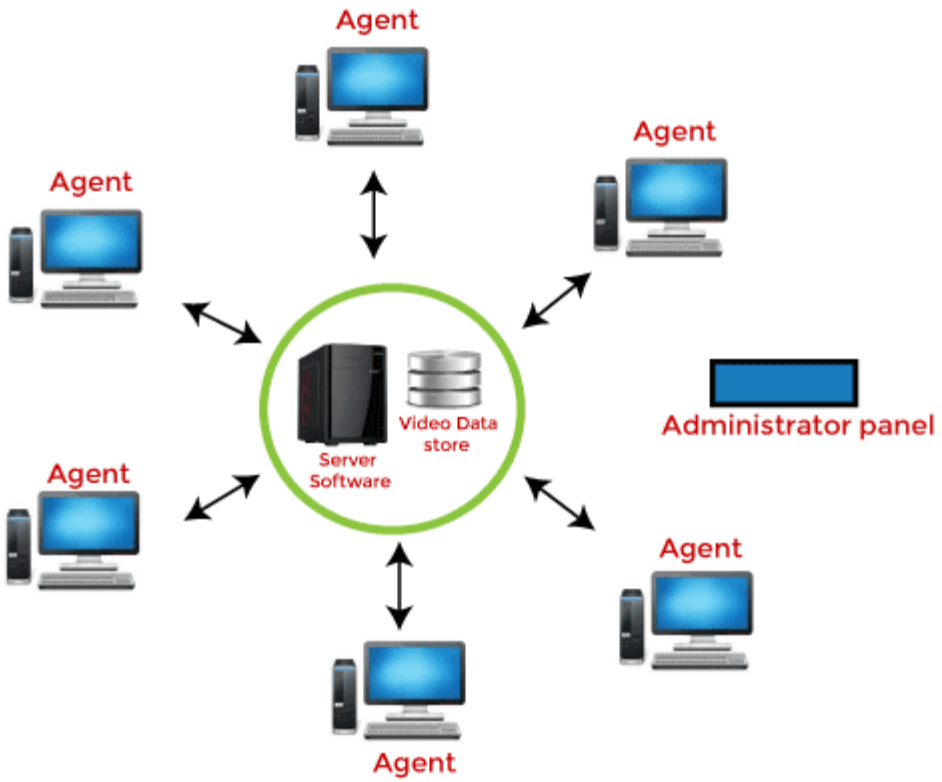
Multitasking Operating System

Advantages of Multitasking OS

- This operating system is more suited to supporting multiple users simultaneously.
- The multitasking operating systems have well-defined memory management.

Disadvantages of Multitasking OS

- The multiple processors are busier at the same time to complete any task in a multitasking environment, so the CPU generates more heat.



An Operating system, which includes software and associated protocols to communicate with other computers via a network conveniently and cost-effectively, is called Network Operating System.

Types of Network operating system are

- Peer to Peer System
- Client Server System

Network Operating System

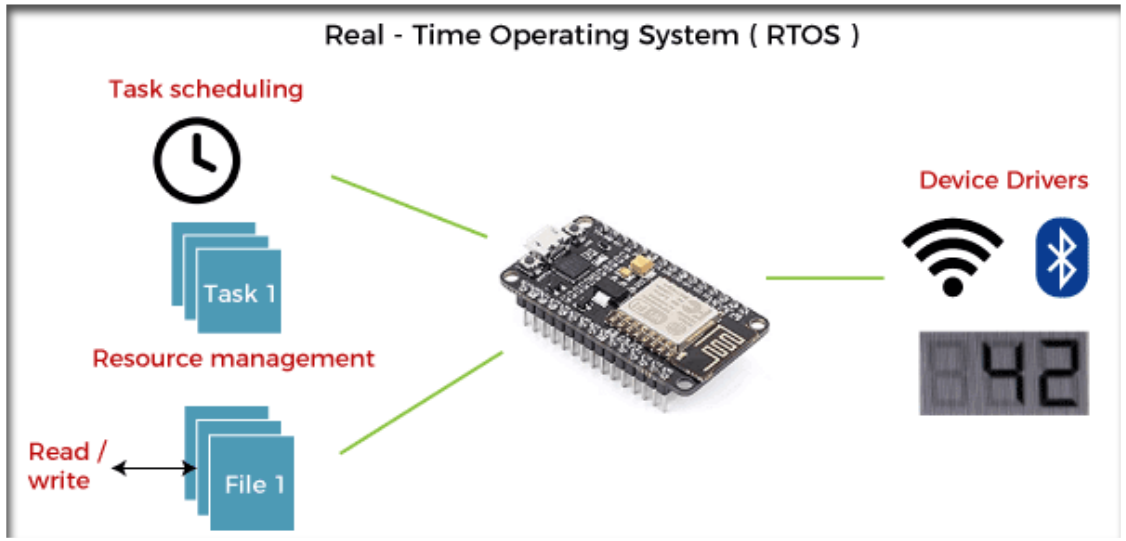
Advantages of Network OS

- In this type of operating system, network traffic reduces due to the division between clients and the server.
- This type of system is less expensive to set up and maintain.

Disadvantages of Network OS

- In this type of operating system, the failure of any node in a system affects the whole system.
- Security and performance are important issues. So trained network administrators are required for network administration.

A Real-Time Operating System (RTOS) is designed to handle time-sensitive tasks where processing must occur within a strict deadline.



In Real-Time Systems, each job carries a certain deadline within which the job is supposed to be completed, otherwise, the huge loss will be there, or even if the result is produced, it will be completely useless. The Application of a Real-Time system exists in the case of military applications, if you want to drop a missile, then the missile is supposed to be dropped with a certain precision.

Types of Real Time operating system are

- Hard Real Time operating system
- Soft Real Time operating system
- Firm Real Time operating system

Real Time Operating System

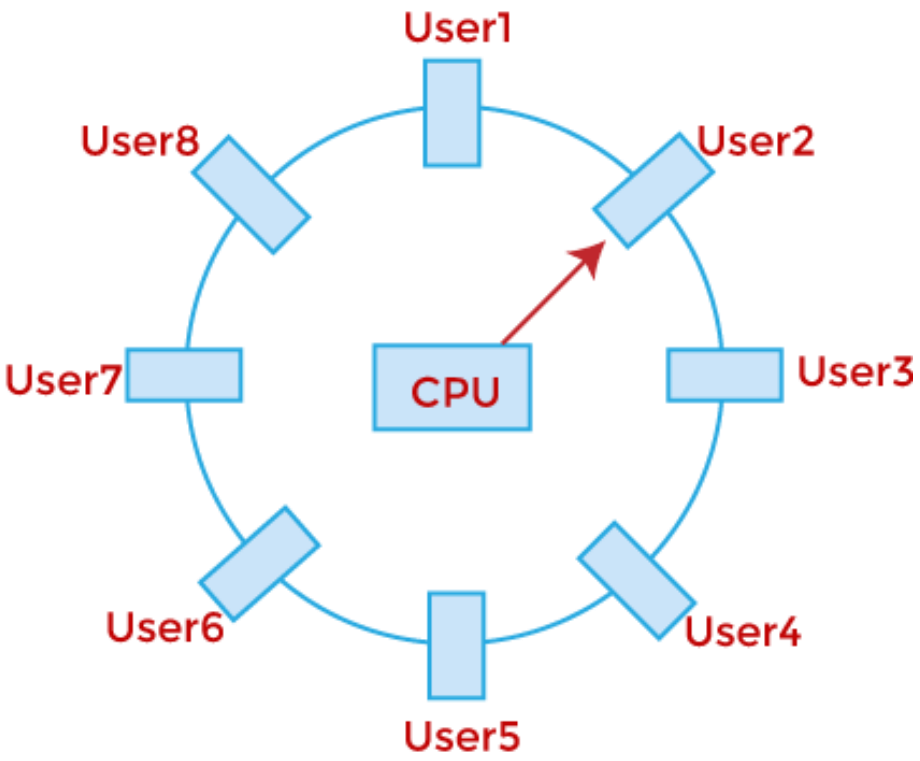
Advantages of RTOS

- Reduced downtime
- Task Administration
- Availability
- Dependability

Disadvantages of RTOS

- Multitasking
- Driver request — Device drivers must be highly optimized for RTOS; otherwise, they may cause unexpected delays.
- Program crashes — Since RTOS prioritizes real-time execution, a single program failure can crash the entire system.
- Task focus

Time Sharing Operating System



Timesharing in case of 8 users

In the Time Sharing operating system, computer resources are allocated in a time-dependent fashion to several programs simultaneously. Thus it helps to provide a large number of user's direct access to the main computer. It is a logical extension of multiprogramming. In time-sharing, the CPU is switched among multiple programs given by different users on a scheduled basis.

A time-sharing operating system allows many users to be served simultaneously, so sophisticated CPU scheduling schemes and Input/output management are required.

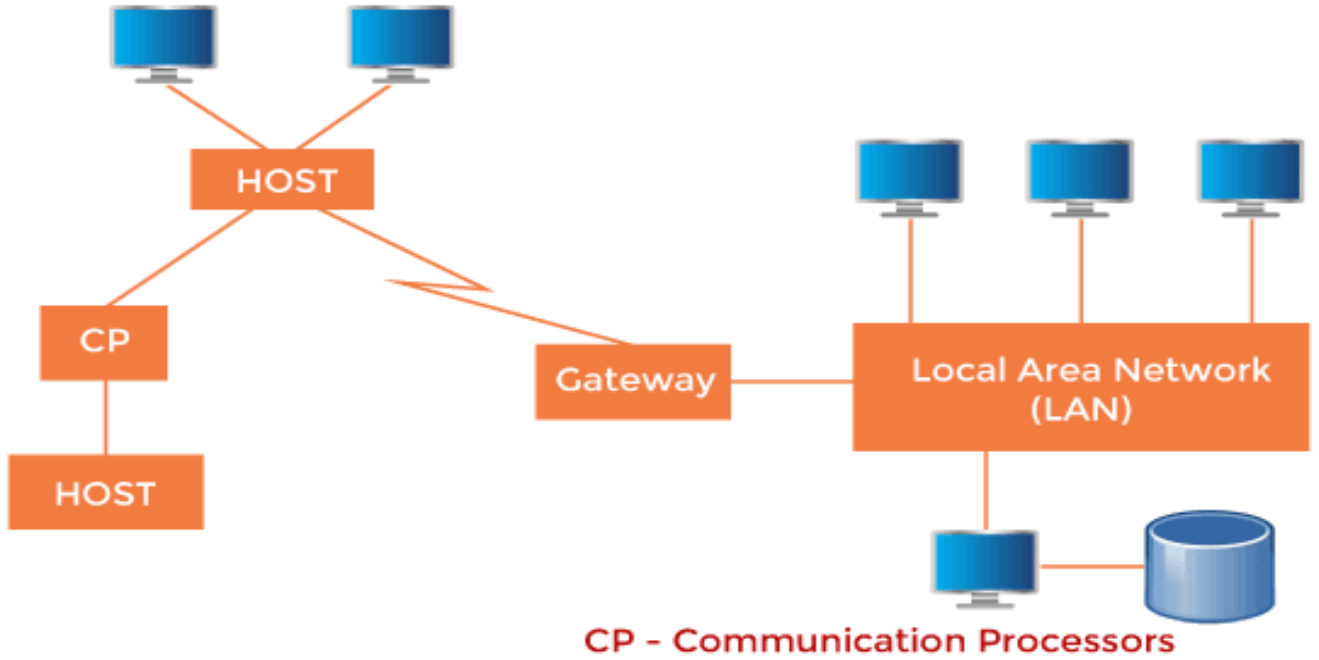
Advantages of Time Sharing OS

- The time-sharing operating system provides effective utilization and sharing of resources.
- This system reduces CPU idle and response time.

Disadvantages of Time Sharing OS

- Data transmission rates are very high in comparison to other methods.
- Security and integrity of user programs loaded in memory and data need to be maintained as many users access the system at the same time.

The Distributed Operating system is not installed on a single machine, it is divided into parts, and these parts are loaded on different machines. A part of the distributed Operating system is installed on each machine to make their communication possible. Distributed Operating systems are much more complex, large, and sophisticated than Network operating systems because they also have to take care of varying networking protocols.



A Typical View of a Distributed System

Distributed Operating System

Advantages of Distributed OS

- The distributed operating system provides sharing of resources.
- This type of system is fault-tolerant.

Disadvantages of Distributed OS

- Protocol overhead can dominate computation cost.

What is a System Call

A computer operates in two modes: User mode and Kernel mode. The execution of a program is in the user mode. When the program requires RAM or a hardware resource, it sends a request to the kernel and the mode changes from user mode to kernel mode. System calls refer to these requests. After completing the task, the mode changes back to the user mode. Likewise, the mode changes continuously. This transition is also called context switching.

System Calls

shell: acts as an intermediary b/w user and system call. Interpret user command and makes system calls.

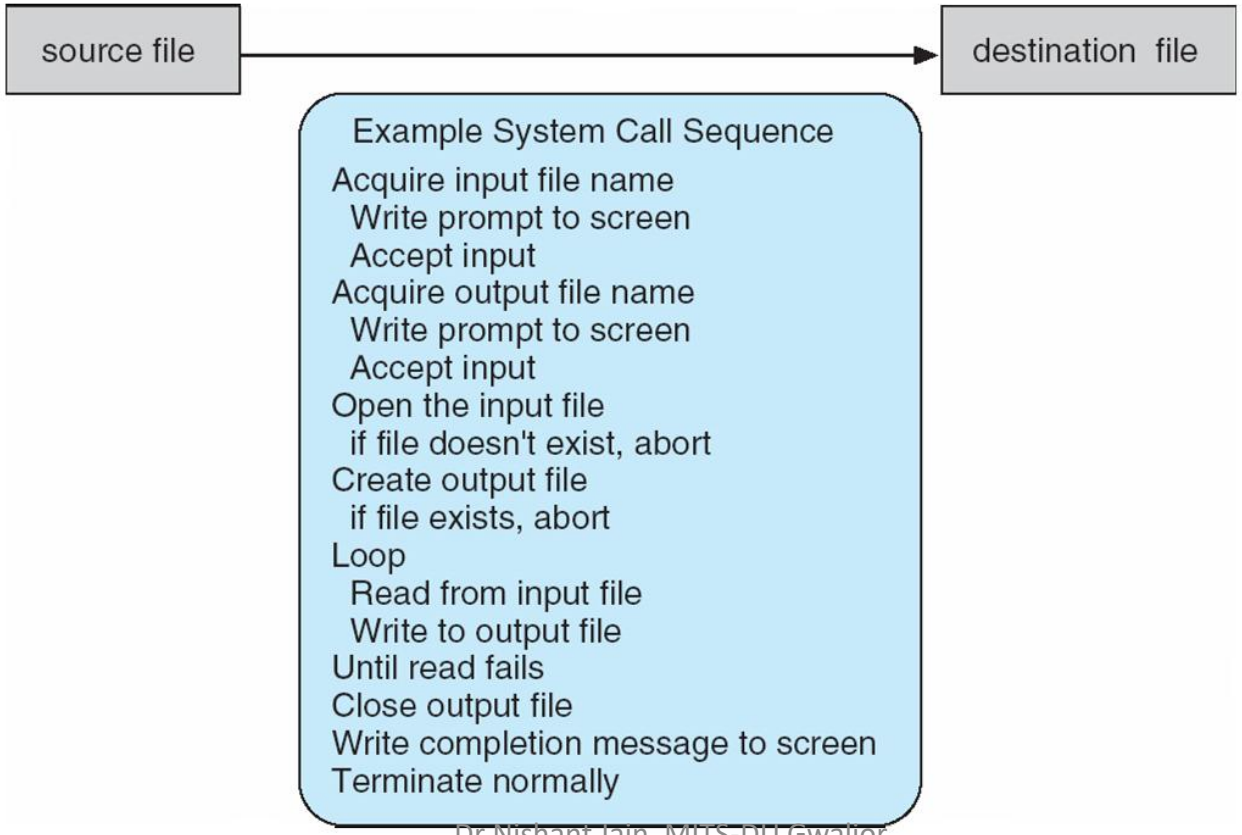
- A system call is a method for a computer program to request a service from the kernel of the operating system on which it is running.
- A system call is a method of interacting with the operating system via programs.
- A system call is a request from computer software to an operating system's kernel.
- The **Application Program Interface (API)** connects the operating system's functions to user programs.
- It acts as a link between the operating system and a process, allowing user-level programs to request operating system services.
- The kernel system can only be accessed using system calls. System calls are required for any programs that use resources.

System Calls

- Programming interface to the services provided by the OS
- Typically written in a high-level language (C or C++)
- Mostly accessed by programs via a high-level Application Program Interface (API) rather than direct system call use
- Three most common APIs are Win32 API for Windows, POSIX API for POSIX-based systems (including virtually all versions of UNIX, Linux, and Mac OS X), and Java API for the Java virtual machine (JVM)
- Why use APIs rather than system calls?

Example of System Calls

System call sequence to copy the contents of one file to another file



System Program

System programs, also known as **system utilities**, provide a convenient environment for program development and execution. Some of them are simply user interfaces to system calls. Others are considerably more complex. They can be divided into these categories:

- **File modification**. Several text editors may be available to create and modify the content of files stored on disk or other storage devices.
- **Programming-language support**. Compilers, assemblers, debuggers, and interpreters for common programming languages (such as C, C++, Java, and PERL).
- **Program loading and execution**. Once a program is assembled or compiled, it must be loaded (loaders) into memory to be executed. Debugging systems for either higher-level languages or machine language are needed as well.

System Program

Communications. These programs provide the mechanism for creating virtual connections among processes, users, and computer systems. They allow users to send messages to one another's screens, to browse Web pages, to send e-mail messages, to log in remotely, or to transfer files from one machine to another.

Background services. All general-purpose systems have methods for launching certain system-program processes at boot time. Some of these processes terminate after completing their tasks, while others continue to run until the system is halted. Constantly running system-program processes are known as **services**, **subsystems**, or daemons. Example, the network daemon process schedulers, system error monitoring services, print servers, etc.